

Recursion Tree Method, Master Method

$$T(n) = \begin{cases} 1 & \text{if } n = 1 \\ aT(\frac{n}{b}) + n & \text{otherwise} \end{cases} \quad \text{where } \underline{\underline{0 < a < b}}$$

$$T(n) = aT(\frac{n}{b}) + n = a \cdot a \cdot T(\frac{n}{b^2}) + a \cdot \frac{n}{b} + n$$

$$= a^2 \cdot T(\frac{n}{b^2}) + (\frac{a}{b})n + n = a^2 \left[aT(\frac{n}{b^3}) + \frac{n}{b^2} \right] + (\frac{a}{b})n + n$$

$$= a^3 T(\frac{n}{b^3}) + (\frac{a}{b})^2 n + (\frac{a}{b})n + n$$

Continuing this process we get

$$\begin{aligned}
 T(n) &= a^k T\left(\frac{n}{b^k}\right) + \left(\frac{a}{b}\right)^{k-1} n + \left(\frac{a}{b}\right)^{k-2} n + \dots + \left(\frac{a}{b}\right)^1 n + n \\
 &= a^k T\left(\frac{n}{b^k}\right) + n \left[1 + \left(\frac{a}{b}\right) + \left(\frac{a}{b}\right)^2 + \dots + \left(\frac{a}{b}\right)^{k-1} \right]
 \end{aligned}$$

Let us suppose that, $n = b^k$

$$T(n) = a^k \cdot T(1) + n \left[1 + \left(\frac{a}{b}\right) + \dots + \left(\frac{a}{b}\right)^{k-1} \right]$$

$$= a^k T(1) + n \left[\frac{1 - \left(\frac{a}{b}\right)^k}{1 - \frac{a}{b}} \right] = a^k \cdot T(1) + n \left(\frac{b}{b-a} \right) \left(1 - \frac{a^k}{b^k} \right)$$

$$= a^k \cdot 1 + n \left(\frac{b}{b-a} \right) \left(\frac{b^k - a^k}{b^k} \right)$$

$$= a^k + n \left(\frac{b}{b-a} \right) \left(\frac{b^k - a^k}{n} \right) \quad \because n = b^k$$

$$= a^k + \left(\frac{b}{b-a}\right)(b^k - a^k)$$

$$= a^k + \frac{b}{b-a} \cdot b^k - \frac{b}{b-a} \cdot a^k$$

$$= \frac{b}{b-a} \cdot b^k + \left(\frac{b-a-b}{b-a}\right) a^k$$

$$\begin{aligned} T(n) &= \left(\frac{b}{b-a}\right) \cdot b^k - \frac{a}{b-a} \cdot a^{\log_b^n} \\ &= \frac{b}{b-a} \cdot b^k - \frac{a}{b-a} \cdot n^{\log_b^a} \quad \text{let } \alpha = \log_b^a \end{aligned}$$

$$\because a < b \Rightarrow \log_b^a < 1 \Rightarrow 0 < \alpha < 1$$

$$\Rightarrow T(n) = \frac{b}{b-a} \cdot n - \frac{a}{b-a} \cdot n^\alpha \Rightarrow T(n) \in \theta(n)$$

$$T(n) = \begin{cases} c & n = 1 \\ 2T\left(\frac{n}{2}\right) + c & n > 1 \end{cases}$$

- $T(n) =$

$$\begin{aligned}
 &2T(n/2) + c \\
 &2(2T(n/2/2) + c) + c \\
 &2^2T(n/2^2) + 2c + c \\
 &2^2(2T(n/2^2/2) + c) + 3c \\
 &2^3T(n/2^3) + 4c + 3c \\
 &2^3T(n/2^3) + 7c \\
 &2^3(2T(n/2^3/2) + c) + 7c \\
 &2^4T(n/2^4) + 15c \\
 &\dots \\
 &2^kT(n/2^k) + (2^k - 1)c
 \end{aligned}$$

$$T(n) = \begin{cases} c & n = 1 \\ 2T\left(\frac{n}{2}\right) + c & n > 1 \end{cases}$$

- So far for $n > 2^k$ we have
 - $T(n) = 2^k T(n/2^k) + (2^k - 1)c$
- What if $k = \lg n$? (i.e., $n = 2^k$)
 - $T(n) = 2^{\lg n} T(n/2^{\lg n}) + (2^{\lg n} - 1)c$
 $= n T(n/n) + (n - 1)c$
 $= n T(1) + (n-1)c$
 $= nc + (n-1)c = (2n - 1)c$

$$T(n) = \begin{cases} c & n = 1 \\ aT\left(\frac{n}{b}\right) + cn & n > 1 \end{cases}$$

- $T(n) =$
 - $aT(n/b) + cn$
 - $a(aT(n/b/b) + cn/b) + cn$
 - $a^2T(n/b^2) + cna/b + cn$
 - $a^2T(n/b^2) + cn(a/b + 1)$
 - $a^2(aT(n/b^2/b) + cn/b^2) + cn(a/b + 1)$
 - $a^3T(n/b^3) + cn(a^2/b^2) + cn(a/b + 1)$
 - $a^3T(n/b^3) + cn(a^2/b^2 + a/b + 1)$
 - ...
 - $a^kT(n/b^k) + cn(a^{k-1}/b^{k-1} + a^{k-2}/b^{k-2} + \dots + a^2/b^2 + a/b + 1)$

$$T(n) = \begin{cases} c & n = 1 \\ aT\left(\frac{n}{b}\right) + cn & n > 1 \end{cases}$$

■ So we have

■ $T(n) = a^k T(n/b^k) + cn(a^{k-1}/b^{k-1} + \dots + a^2/b^2 + a/b + 1)$

■ For $k = \log_b n$

■ $n = b^k$

■
$$\begin{aligned} T(n) &= a^k T(1) + cn(a^{k-1}/b^{k-1} + \dots + a^2/b^2 + a/b + 1) \\ &= a^k c + cn(a^{k-1}/b^{k-1} + \dots + a^2/b^2 + a/b + 1) \\ &= ca^k + cn(a^{k-1}/b^{k-1} + \dots + a^2/b^2 + a/b + 1) \\ &= cna^k/b^k + cn(a^{k-1}/b^{k-1} + \dots + a^2/b^2 + a/b + 1) \\ &= cn(a^k/b^k + \dots + a^2/b^2 + a/b + 1) \end{aligned}$$